Lecture 16: Q-Theory of Investment

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2026

The Neoclassical Model

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- Discounts future profits at rate r
- Produces a homogeneous good
- Perfect competition in input and output markets
- Production Function is CRS
- Hires labor, buys capital

Problem of a firm that takes input prices as given

Profits

Profits are given by revenues minus the wage bill minus capital expenditures

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- Abstracting from taxes, bonus depreciation, etc.
- Perfect foresight

The firm’s problem

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$$\mathcal{H} = e^{-rt} \left(AF(k_t, n_t) - w_t N_t - i_t \right) + \lambda_t i_t. \tag{6}$$

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Question: What is the meaning of λ_t?

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Optimality conditions:

$$AF_n = w_t \tag{9}$$

$$e^{-rt} = \lambda_t \tag{10}$$

$$e^{-rt} A_t F_k = -\dot{\lambda}_t \tag{11}$$

$$\lim_{t\to\infty} \lambda_t k_t = 0. \tag{12}$$

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$$AF_n = w_t \tag{14}$$

$$q_t = 1 \tag{15}$$

$$AF_k = -\dot{q}_t + r q_t \tag{16}$$

$$\lim_{t\to\infty} e^{-rt} q_t k_t = 0. \tag{17}$$

Zoom in the key result

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- The marginal cost of investing is losing 1 unit of profits
- Optimality requires to invest until marginal benefit equals marginal cost

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- There is no role for past variables

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- Capital should *jump* instantaneously
- Therefore, investment needs to be infinite for an instant

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Imagine A increases to A_H and goes down to A in the following instant

- Firms will infinitely invest in the present

The economics of the basic problem

Imagine A increases to A_H and goes down to A in the following instant

- Firms will infinitely invest in the present
- Firms will infinitely disinvest in the following instant

Against any possible intuition. $r = AF_k$

Away from continuous time for a second

You may think this are problems that arise only in the continuous time limit

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- Assume that $F(K, L) = K^{\alpha} L^{1-\alpha}$. Therefore:

$$1 + r_t = 1 + A_t \alpha K_t^{\alpha-1} L_t^{1-\alpha}$$

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- Make a log-linear approximation. Hatted variables are log changes:

$$\hat{r}_t = \frac{r}{1+r}\left(\hat{a}_t - (1-\alpha)\hat{k}_t + (1-\alpha)\hat{l}_t\right)$$

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- where $\hat{r}_t = \log \frac{1+r_t}{1+r}$, solve for $\hat{k}$

$$\hat{k}_t = -\hat{r}_t \left(\frac{1+r}{(1-\alpha)r}\right) + \frac{\hat{a}_t}{1-\alpha} + \hat{l}_t$$

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- Assume 100% of GDP could be transformed to capital.
- Capital-output ratios are between 2 and 4 (depending on land and housing)
- To increase capital by 31%, it would take 61%-124% of GDP
- With δ: to increase capital by 14%, it would take 28-56% of GDP

The economics of the basic problem

Imagine firms notice that A will increase at time τ ($A_\tau > A$)

- Firms do nothing up until period τ
- invest $+\infty$ at time τ

No role for news about the future

Idea: convex adjustment costs

Investing takes time and resources

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Convex costs

$$\phi(i,k) = \frac{\varphi}{2}\left(\frac{i}{k}\right)^2 k \tag{18}$$

Lucas (1967) is an early reference for convex costs on investment

- Note that defined in this way $\phi(i,k)$ is homogeneous of degree 1

New firm's problem

Objective

$$\max_{i_t, n_t} V = \int_0^\infty e^{-rt} \pi_t dt \tag{19}$$

Subject to

$$\pi_t = A_t F(k_t, n_t) - w_t n_t - i_t - \frac{\varphi}{2} \left(\frac{i}{k} \right)^2 k \tag{20}$$

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Question: What is the meaning of λ_t?

Optimality Conditions

$$AF_N = w_t \tag{23}$$

$$1 + \varphi \frac{i_t}{k_t} = q_t \tag{24}$$

$$-\dot{q}_t + rq_t = AF_k + \frac{\varphi}{2}\left(\frac{i}{k}\right)^2 \tag{25}$$

Zoom in the important economics

$$1 + \varphi \frac{i_t}{k_t} = q_t$$

- Rewrite the equation as

$$\frac{i_t}{k_t} = h(q_t)$$

- The investment rate is **only** a function of q_t

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- q is a *sufficient statistic*

Two benefits for having more capital

- You can solve the differential equation

$$-\dot{q}_t + rq_t = AF_k + \frac{\varphi}{2}\left(\frac{i}{k}\right)^2$$

- And find that

$$q_0 = \int_0^\infty e^{-rt}\left[A_t F_{kt} + \frac{\varphi}{2}\left(\frac{i_t}{k_t}\right)^2\right] dt$$

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- Reductions in future costs of adjustment
- that is the marginal value of investment.

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- a higher q calls for higher investment

$$\frac{i_t}{k_t} = \frac{q_t - 1}{\varphi}$$

- Higher k brings down future MPK
- Lowering q

When to invest

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$$\frac{i_t}{k_t} = \frac{q_t - 1}{\varphi}$$

- Investment will be positive as long as $q > 1$
- How much to invest depends on φ
- What happens in the limit $\varphi \to 0$?

How to implement the Q Theory

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- We do not have good measures for marginal q in the data
- It is a similar measurement problem to that of having measures of marginal cost!
- We do have measures of *average* q. Also called *Tobin's* Q
- To which we will refer to as Q.

Tobin's Q

- Tobin (1969) argued that firms should invest if

$$Q = \frac{\text{Value of the Firm Capital}}{\text{Replacement Cost of the Capital Stock}}$$

- Capital is valued more within the firm, than outside of the firm.

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- Under what conditions Tobin's claim is sustained in our standard model?

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- Average Q is the average value of a unit of capital

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$$Q = \frac{\text{Value of the Firm Capital}}{\text{Replacement Cost of the Capital Stock}}$$

- Can we learn about q with information about Q?

Hayashi 1982

- Under a number of conditions, there is an equivalent result.
  - Firms have CRS technology
  - The adjustment cost technology is CRS
  - Perfect competition
  - Efficient Financial Markets
- Then $q_t = Q_t$
- Hayashi (1982) presented this result in continuous time.
- I will show you the result in discrete time, which for me is more intuitive

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$$\pi_t = AF(K_t, L_t) - W_t L_t - I_t - \psi\left(\frac{I_t}{K_t}\right) K_t$$

- Firms maximize their value

$$V_0 = \sum_{t=0}^{\infty} \left(\frac{1}{1+r}\right)^t \pi_t$$

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- Subject to the law of motion of capital

$$K_{t+1} = K_t(1-\delta) + I_t$$

- q_t is the Lagrange multiplier

Marginal q and Average Q in discrete time

- The problem has very similar first order conditions

$$AF_L(K_t, L_t) = W_t$$

$$q_t = 1 + \psi'\left(\frac{I_t}{K_t}\right)$$

$$q_t = \frac{1-\delta}{1+r} q_{t+1} + \frac{1}{1+r}\left(A_{t+1}F_k(K_{t+1}, L_{t+1}) - \psi\left(\frac{I_{t+1}}{K_{t+1}}\right) - \psi'\left(\frac{I_{t+1}}{K_{t+1}}\right)\frac{I_{t+1}}{K_{t+1}}\right)$$

Marginal q and Average Q in discrete time

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- Use the law of motion of capital in period $t+1$ to replace I_{t+1}

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Marginal q and Average Q in discrete time

- Rearrange

$$q_t K_{t+1} = \frac{1}{1+r} q_{t+1} K_{t+2} + \frac{1}{1+r} \left(A_{t+1} F_k(K_{t+1}, L_{t+1}) K_{t+1} - \psi\left(\frac{I_{t+1}}{K_{t+1}}\right) K_{t+1} - I_{t+1} \right)$$

- Now use Euler's Theorem $F(K,L) = F_k(K,L)K + F_L(K,L)L$
- And, perfect competition in the labor market $W = AF_L(K,L)$

$$q_t K_{t+1} = \frac{1}{1+r} q_{t+1} K_{t+2} + \frac{1}{1+r} \left(A_{t+1} F(K_{t+1}, L_{t+1}) - W_{t+1} L_{t+1} - \psi\left(\frac{I_{t+1}}{K_{t+1}}\right) K_{t+1} - I_{t+1} \right)$$

- Notice that the term in parenthesis is just profits tomorrow

$$q_t K_{t+1} = \frac{1}{1+r} q_{t+1} K_{t+2} + \frac{1}{1+r} \pi_{t+1}$$

Marginal q and Average Q in discrete time

- Iterate forward and use the TVC to get rid of the terminal term

$$q_t K_{t+1} = \sum_{j=1}^{\infty} \left(\frac{1}{1+r} \right)^t \pi_{t+1}$$

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$$q_t K_{t+1} = V_t^e$$

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- Realize that the RHS is the average Q

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- Why do we want to test this theory?
  - It is our benchmark model
  - Many educated guesses come from this framework
  - The investment effects of tax reforms
  - The investment effects of monetary policy
  - And so on...